MILITARY SPECIFICATION

AMPLIFIER-MIXER
AM-7157/DKT-59

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, development and test requirements for the AM-7157/DKT-59, Amplifier-Mixer, critical item, referred to herein as the amplifier.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, New Jersey 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5815

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-T-85650	Transmitting Set, Telemetric Data AN/DKT-59
MIL-T-85653	Transmitter, Radio T-1458/DKT-59

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-2078	Requirements for Preparation of Support Equipment Depot Level Rework Standards
MIL-STD-45662	Calibration System Requirements

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

1356AS312	Amplifier-Mixer

(Application for copies should be addressed to the Department of Defense Technical Information Center, Cameron Station, Alexandria, VA 22314.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Amplifier description. The amplifier shall mix and amplify control of the composite signal of the subcarrier oscillator group containing continuous information. The output will be used to frequency modulate the RF carrier of the radio transmitter module, T-1458/DKT-59, MIL-T-85653(AS).

3.1.1 Interface definition. The amplifier's physical and electrical interfaces with the module assembly unit shall be in accordance with Drawing 1356AS312.

3.2 Characteristics.

3.2.1 Performance. The amplifier shall meet the performance requirements specified herein under the conditions of 3.2.5.

3.2.1.1 External power. The amplifier shall operate using external power with characteristics as specified in MIL-STD-704 and modified herein as follows:

a. 20.0 to 36.0 volts direct current (Vdc) (see 4.6.2.8).

b. Transient surge from +40 to -40 Vdc (see 4.6.2.8.2).

c. Maximum current shall be not greater than 0.010 amperes (see 4.6.2.8.1.a).

3.2.1.2 Input characteristics. The amplifier shall have the following input characteristics (4.6.2.8.1):

a. Input impedance: 9.5 to 10.5 Kilohms (K ohms)

b. Input signal: 0.1 to 0.6 volts root mean square (Vrms)

c. Input frequency: 0.300 to 200 Kilohertz (KHz)

3.2.1.3 Output characteristics. The amplifier output characteristics shall meet the requirements as follows:

a. Gain. The gain shall be fixed at 3.1 ±10 percent with a supply voltage variation from 20.0 to 36.0 Vdc and shall be flat to ±0.5 dB from 0.300 to 200 KHz (see 4.6.2.2).

b. Impedance. The amplifier output impedance shall be not greater than 500 ohms (see 4.6.2.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.2.1.4 Linearity. The linearity of the output voltage waveform at the output of the amplifier shall be an exact amplified duplicate of the sub-carrier composite input voltage waveform. There shall be no limiting, flat spots, or noise spikes, and the positive half cycle shall be equal in magnitude to the negative half cycle to within ±1.0 percent for all input voltage variations (see 4.6.2.4).

3.2.1.5 Harmonic distortion. The harmonic distortion on the output of the amplifier shall be not greater than 0.5 percent for all input voltage variations (see 4.6.2.5).

3.2.1.6 Ripple and noise on output. The ripple and noise on the output of the amplifier shall be not greater than 0.001 volts rms for all input voltage variations (see 4.6.2.6).

3.2.1.7 Short circuit protection. The amplifier shall recover automatically to specified performance with no effect on reliability following removal of a short circuit condition (see 4.6.2.7).

3.2.1.8 Transient surge voltage protection. The amplifier shall recover automatically to specified performance with no effect on reliability following a transient surge dc voltage step function defined in 3.2.1.1 see 4.6.2.8.2).

3.2.2 Physical characteristics. The amplifier shall meet the physical requirements of Drawing 1356AS312 (see 4.6.1.1).

3.2.3 Reliability. The reliability of the amplifier shall be not less than 2000 hours mean-time-between-failures (MTBF) (see 4.6.4).

3.2.4 Maintainability. The maintainability program, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in MIL-STD-2078.

3.2.5 Environmental conditions.

3.2.5.1 Operational. The amplifier shall meet all requirements of 3.2.1 and 3.2.3 prior to, during, and after exposure to the following environmental conditions, separately or any combination thereof. The amplifier shall not exhibit any degradation in excess of limits specified herein as the result of exposure to any of the environments in a through f of Table I (see 4.6.3 and 4.6.3.1).

3.2.5.2 Non-operational. The amplifier shall meet all requirements of 3.2.1 and 3.2.3 after exposure to the environmental conditions c, g, and h of Table I, separately or any combination thereof. The amplifier shall not

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

exhibit any temporary or permanent degradation in excess of limits specified herein as the result of exposure to any of the environments specified above (see 4.6.3.2).

TABLE I. Environmental conditions.

PARAMETER	OPERATIONAL	NON-OPERATIONAL
a. Pressure	70,000 ft (equivalent pressure)	-----
b. High Temperature	+71°C	-----
c. Low Temperature	-40°C	-40°C
d. Acceleration	40 g, 3 axes	-----
e. Random Vibration	20-2000 Hz, .2g^2/Hz	-----
f. Shock	65g, 5ms, 1/2 sine	-----
g. Thermal Shock	-----	-55°C to +85°C
h. Moisture Resistance	-----	95% relative humidity

3.2.6 Storage life. The storage life of the amplifier shall be such that it can be stored for three years in the shipping or storage container and then meet the requirements specified herein.

3.3 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3.1 Visual examinations. Test samples shall be subjected to visual examination to verify conformance to the physical and interchangeability requirements of this specification and the contract (see 4.6.1).

3.4 Design and construction. Design and construction shall be in accordance with Drawing 1356AS312 and requirements specified herein (see 3.4.1 and 4.6.1.9).

3.4.1 Materials, processes, and parts. Fulfillment of major design objectives for the intended application specified herein shall be the prime consideration in the selection of materials, processes, and parts. In so doing, the following shall govern:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein (see 4.6.1.2).

b. Other materials, processes, and parts requirements shall conform to MIL-E-5400 (see 4.6.1.2).

c. Parts shall be selected using the following order of preference (see 4.6.1.2).

(1) Parts listed on a Qualified Parts List (QPL) with two or more sources.

(2) Parts listed on a QPL with only one source.

(3) Commercial parts with two or more sources.

(4) Commercial parts with only one source.

3.4.1.1 Standard parts. Parts specified in 3.4.1c (1) and (2) are standard parts and do not require approval from the procuring activity. Parts selected for use from 3.4.1c (3) and (4) are considered nonstandard and require approval. Nonstandard parts approval requests shall be submitted on DD Form 2052 in accordance with the contract (see 6.2.2).

3.4.1.2 Commercial parts. Commercial parts specified in 3.4.1c (3) and (4) shall be readily available, nationally catalogued, off-the-shelf items.

3.4.1.3 Nonstandard parts. Use of nonstandard parts in first article or production units prior to government approval is at the sole risk of the contractor and subject to replacement if the government does not approve the proposed nonstandard part.

3.4.2 Electromagnetic interference.

The amplifier shall meet the electromagnetic interference (EMI) requirements of MIL-STD-461, tests CE03, CE07, CE01, CS01, CS02, CS06, RE02, RS02, and RS03.

3.4.3 Nameplate and marking. The nameplate shall be as specified in Drawing 1356AS312. Interchangeable parts and assemblies shall be legibly marked. All markings shall be as specified in MIL-STD-130 (see 4.6.1.3).

3.4.4 Serial numbers. Serial number assignments shall be as specified in MIL-N-18307 (see 4.6.1.3 and 6.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.5 Interchangeability. All assembly and subassembly level items of hardware that are subject to replacement and have the same part number shall be electrically, physically, and functionally interchangeable. Electrical component parts shall conform to the requirements specified in MIL-STD-965. The amplifier shall meet the interchangeability requirements of MIL-STD-454, Requirement 7, and the electrical and mechanical interface requirements of the AN/DKT-59 Telemetric Data Transmitting Set (MIL-T-85650) (see 4.6.1.6 and 4.6.2.9).

3.4.6 Safety. The amplifier shall meet the safety requirements of MIL-STD-454, Requirement 1 (see 4.6.1.7).

3.4.7 Human engineering. All controls and adjustments shall be set during production and initial testing and shall be inaccessible in the field (see 4.6.1.8).

3.5 Soldering. Soldering practices shall be in accordance with MIL-STD-454, Requirement 5 (see 4.6.1.4).

3.6 Workmanship. Fabrication and workmanship practices shall be in accordance with MIL-STD-454, Requirement 9 (see 4.6.1.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to monitor any of the inspections or to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall assure product conformance to the requirements, inspections and tests specified herein. The contractor's Quality Assurance Program shall be planned and conducted in a manner to achieve levels of quality and reliability commensurate with the intended application specified herein.

4.1.1.1 Inspection. The contractor's inspection system shall conform to MIL-STD-883 for both the commercial off-the-shelf microelectronics and new contractor-designed (peculiar) microelectronics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.1.1.2 Test equipment and inspection facilities. The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.1.2 Test plans and procedures. Detailed test plans and procedures in accordance with MIL-T-18303 shall be prepared by the contractor (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract (see 6.2.1). When approved test procedures are available from previous contracts, such procedures will be provided and may be used when approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2.1).

4.1.3 Test data. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare test data showing quantitative results for all first article and quality conformance inspections (see 6.2.2.) for each amplifier offered for acceptance. This report shall also identify all rejections from tests.

4.1.3.1 Electromagnetic interference test data. The contractor shall prepare a electomagnetic interference test report in accordance with MIL-STD-831 (see 6.2.2).

4.1.4 Failure reports. All failures shall be documented in a failure report, identifying the cause of failure and the proposed corrective action for each failure (see 6.2.2).

4.1.5 Reliability Reports. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare reliability test data, including failure reports, in accordance with MIL-STD-781 (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

(1) Visual examination (see 4.6.1)

(2) Performance test (see 4.6.2)

(3) Environmental test (see 4.6.3)

(4) Reliability test (see 4.6.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

b. Quality conformance inspection (see 4.5)

(1) Individual test (see 4.5.2)

(2) Periodic production test (see 4.5.3)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.3.3.

4.3.1 Maintenance and calibration. All test equipment shall be calibrated and maintained in accordance with MIL-STD-45662. Certification to this effect shall accompany all reports of first article and quality conformance tests (see 6.2.2).

4.3.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used shall have an accuracy within 20 percent of the tolerance for the variable to be measured. In those cases where an open-ended minimum or maximum requirement is given, the test equipment used shall be accurate to within five percent of that requirement.

4.3.3 Test environment. Unless otherwise specified in the applicable test procedure, all tests and inspections shall be conducted under the following environmental conditions:

a. Temperature (ambient): +23 ±10°C

b. Altitude: elevation of test facility

c. Relative humidity: 50 ±30 percent

4.3.4 Tolerances on environmental test conditions. The allowable tolerances on environmental test conditions shall be in accordance with MIL-T-5422.

4.4 First article inspections. When specified in the contract (see 6.2.1), first article inspections shall be conducted by the contractor on an amplifier representative of the production amplifiers to be supplied under the contract. The first article sample shall be subjected to the examinations and tests of Table II and shall be performed in the sequence specified therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE II. Individual tests.

Examination or Test*	Requirement Paragraph	Method Paragraph		
		First Article	Quality Conformance	
			Individual	Periodic Production
Visual examination	3.3.1	4.6.1	4.6.1	4.6.1
Performance tests				
Electromagnetic interference	3.4.2	4.6.2.1		
Gain	3.2.1.3(a)	4.6.2.2	4.6.2.2	4.6.2.2
Impedance	3.2.1.3(b)	4.6.2.3		
Linearity	3.2.1.4	4.6.2.4		4.6.2.4
Harmonic distortion	3.2.1.5	4.6.2.5	4.6.2.5	4.6.2.5
Ripple and noise on output	3.2.1.6	4.6.2.6	4.6.2.6	4.6.2.6
Short circuit protection	3.2.1.7	4.6.2.7		
External power voltage variation	3.2.1.1	4.6.2.8		4.6.2.8
Steady-state voltage	3.2.1.1	4.6.2.8.1		4.6.2.8.1
Transient surge dc voltage protection	3.2.1.8	4.6.2.8.2	4.6.2.8.2	4.6.2.8.2
Interchangeability	3.4.5	4.6.2.9	4.6.2.9	4.6.2.9
Environmental test	3.2.5	4.6.3		4.6.3
Reliability test	3.2.3	4.6.4		4.6.4

*All examinations and tests shall be performed in top to bottom sequence.

4.4.1 Acceptance criteria. Failure of any amplifier to pass any first article test shall result in discontinuance of testing and rejection of the first article sample. The procuring activity reserves the right to require

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

an additional sample and test if any amplifier fails to meet any requirement of this specification (see 6.2.1).

4.4.1.1 Post-tests. An amplifier which has previously failed first article inspection shall be repaired and retested in accordance with 4.6.2 prior to its resubmission for first article acceptance.

4.4.2 First article failure analysis and corrective action. The procuring activity shall be notified of a first article test failure and its probable cause (see 4.1.4 and 6.2.2).

4.5 Quality conformance inspections. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspections. All inspection and testing shall be subject to monitoring by the Government inspector. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspections shall consist of the following:

a. Individual tests (see 4.5.2)

b. Periodic production tests (see 4.5.3)

4.5.1 Quality conformance acceptance criteria. Any amplifier that fails to pass any individual test shall be rejected.

4.5.2 Individual tests. The individual tests shall be conducted on each amplifier offered for acceptance and shall consist of the tests in Table II.

4.5.3 Periodic production tests. After the first 50 amplifiers are produced, a periodic sample consisting of one unit shall be selected at random from the first 10 amplifiers of every 50 produced thereafter. This sample shall be selected from amplifiers which have previously passed the individual tests of 4.5.2. The sample shall be subjected to all the examinations and tests specified in Table II.

4.5.3.1 Periodic production sample rejection. If the periodic production sample fails to meet the requirement of any examination or test specified herein, the sample shall be rejected. Before any further production lots or amplifiers may be submitted for acceptance, a detailed report shall be prepared (see 6.2.2) describing the rejection and action taken to prevent recurrence of the defect which caused the failure. Further production lots will not be considered for acceptance until the periodic production sample has been approved.

4.6 Examination and test methods. Details of methods and procedures shall be in accordance with the approved test plan (see 4.1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.6.1 Visual examination.

4.6.1.1 Physical characteristics. The amplifier shall be inspected visually for conformance to 3.2.2.

4.6.1.2 Materials, processes, and parts. The amplifier shall be examined visually for conformance to 3.4.1.

4.6.1.3 Nameplates and product marking. The amplifier shall be inspected for conformance to 3.4.3.

4.6.1.4 Soldering. The amplifier shall be examined for conformance to 3.5.

4.6.1.5 Workmanship. The amplifier shall be examined for conformance to 3.6.

4.6.1.6 Interchangeability. The amplifier shall be examined to verify physical interchangeability conformance to 3.4.5.

4.6.1.7 Safety. The amplifier shall be examined to verify conformance to the safety requirements of 3.4.6.

4.6.1.8 Human engineering. The amplifier shall be examined to verify conformance to 3.4.7.

4.6.1.9 Design and construction. The amplifier shall be examined to verify conformance to the requirements of 3.4.

4.6.1.10 Maintainability. The amplifier maintainability program, construction, packaging and access to test points shall be examined to verify conformance to the requirements of 3.2.4.

4.6.2. Performance tests. Performance tests shall be performed as specified herein.

4.6.2.1 Electromagnetic interference. The amplifier sections shall be tested for conformance to 3.4.2 and in accordance with MIL-STD-462.

4.6.2.2 Gain. Apply external power of +24.0 Vdc to pin 5 and a common to pin 2 in accordance with 3.2.1.1. Apply a sine wave test signal with an amplitude range from 0.1 to 0.6 Vrms to pin 6 (pin 2 common) and the test signal frequency shall be varied from 0.300 to 200 KHz in accordance with 3.2.1.2. The output gain at pin 1 (pin 2 common) shall meet the requirements of 3.2.1.3(a).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.6.2.3 Impedance. Repeat the external test method conditions of 4.6.2.2. The output load and output amplitude shall meet the requirements of 3.2.1.3(b).

4.6.2.4 Linearity. Apply external power to pin 5 and a common to pin 2 and apply a load to pin 6 (pin 2 common) in accordance with 3.2.1.1, 3.2.1.3(a), and 3.2.1.3(b). Apply a sine wave test signal with an amplitude range from 0.1 to 0.6 Vrms to pin 6 (pin 2 common) and the test signal frequency shall be varied from 0.300 to 200 KHz in accordance with 3.2.1.2. The linearity at pin 6 (pin 2 common) shall meet the requirements of 3.2.1.4.

4.6.2.5 Harmonic distortion. Apply external power to pin 5 and a common to pin 2 and apply a load to pin 6 (pin 2 common) in accordance with 3.2.1.1, 3.2.1.3(a), and 3.2.1.3(b). Apply a sine wave test signal with an amplitude range from 0.1 to 0.6 Vrms to pin 6 (pin 2 common) and the test signal frequency shall be varied from 0.300 to 200 KHz in accordance with 3.2.1.2 The harmonic distortion at pin 6 (pin 2 common) shall meet the requirements of 3.2.1.5.

4.6.2.6 Ripple and noise on output. Apply external power to pin 5 and a common to pin 2 and apply a load to pin 1 (pin 2 common) in accordance with 3.2.1.1, 3.2.1.3(a), and 3.2.1.3(b). Apply a common to pin 6 (pin 2 common). The ripple and noise at pin 1 (pin 2 common) shall meet the requirements of 3.2.1.6.

4.6.2.7 Short circuit protection. Apply external power to pin 5 and a common to pin 2 in accordance with 3.2.1.1. Apply a direct short to pin 1 and pin 2 (common) in parallel for a period of three minutes. The dead short shall be removed. The output voltage at pin 1 (pin 2 common) shall meet the requirements of 3.2.1.7.

4.6.2.8 External power voltage variation. External power voltage variation shall be as specified in 4.6.2.8.1 and 4.6.2.8.2.

4.6.2.8.1 Steady-state voltage test. External power shall be applied to the amplifier. The external voltage shall be adjusted to +20.0 +0.25,-0.0 Vdc and the following verified:

a. Power demand as specified in 3.2.1.1.

b. Input characteristics as specified in 3.2.1.2.

c. Output characteristics as specified in 3.2.1.3.

d. Linearity as specified in 3.2.1.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

e. Harmonic distortion as specified in 3.2.1.5.

f. Ripple and noise as specified in 3.2.1.6.

With external voltage of +36.0 +0.0,-0.25 Vdc applied, the test shall be repeated.

4.6.2.8.2. Transient surge dc voltage protection. Apply external power to pin 5 and a common to pin 2 in accordance with 3.2.1.1 for a period of 3 minutes. The external voltage shall be reduced to zero for not less than 10 seconds, raised to +40.0, ±1.0 Vdc for not less than 10 seconds, then to -40.0 ±1.0 Vdc for 1.0 second, and returned to a nominal +28.0 Vdc. The output voltage at pin 1 (pin 2 common) shall meet the requirements of 3.2.1.8.

4.6.2.9 Interchangeability. The amplifier shall be tested to verify electrical and operational interchangeability conformance to 3.4.5.

4.6.3 Environmental tests. Environmental tests shall be performed as specified herein. When tests are conducted at high or low temperatures, test amplifiers shall be stabilized at the specified temperature for not less than two hours prior to testing. If the amplifiers are removed from the chamber for any portion of the tests, such tests shall be completed within 10 minutes of removal from the chamber. If the tests are not completed within the 10 minute period, the amplifiers shall be returned to the temperature chamber for not less than two hours before continuing the tests. The test amplifiers shall stabilize at room ambient conditions prior to performance of post tests and measurements.

4.6.3.1 Operational environmental tests. The amplifier performance shall be tested as specified in 4.6.3.1.1 through 4.6.3.1.6 and meet the requirements of 3.2.1.1 through 3.2.1.8 during and after exposure to the environmental conditions specified in Table I, a through f.

4.6.3.1.1 Reduced pressure. The amplifier shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.2 High temperature. The amplifier shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.3 Low temperature. The amplifier shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.4 Acceleration. The amplifier shall be tested in accordance with 3.2.5.1 and Requirement d. of Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.6.3.1.5 Random vibration. The amplifier shall be tested with a spectral power density of 0.2 g^2/Hz from 20 to 2000 Hz for 2 hours in each of three mutually perpendicular axes.

4.6.3.1.6 Shock. The amplifier shall be tested in accordance with MIL-T-5422, Shock Test.

4.6.3.2 Nonoperational environmental tests. The amplifier's performance shall be tested as specified in 4.6.3.2.1 through 4.6.3.2.3 and meet the requirements of 3.2 after exposure to the environmental conditions specified in Table I, c, g, and h.

4.6.3.2.1 Low temperature. The amplifier shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 equipment Class 2 at -40°C for 48 hours.

4.6.3.2.2 Thermal shock. The amplifier shall be tested in accordance with MIL-T-5422, Temperature Shock Test.

4.6.3.2.3 Moisture resistance. The amplifier shall be tested in accordance with MIL-T-5422, Humidity Test.

4.6.4 Reliability test. The amplifier shall be tested as specified in the Probability Ratio Sequential Test (PRST) Plan VIIIC, Figure C-8, Appendix C of MIL-STD-781 to demonstrate compliance with 3.2.3.

4.6.4.1 Reliability acceptance criteria. The Government inspector shall be notified any time the total test hours and test failures plotted in accordance with the PRST Acceptance and Reject criteria for Test Plan VIIIC of MIL-STD-781 reflect a "reject" situation during the conduct of the contractor's approved production test procedures. The procuring activity reserves the right to stop the acceptance of products at any time a reject situation exists pending a review of the contractor's efforts to improve the product, product components and parts and the product workmanship.

4.7 Production amplifiers. Amplifiers supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article. Each amplifier shall be capable of successfully passing the same tests as imposed on the first article. Evidence of noncompliance with the above shall constitute cause for rejection, and, for amplifiers already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections.

4.8 Packaging inspection. The preservation, packaging, packing and marking of the amplifier shall be inspected to verify conformance to the requirements of Section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the level of preservation, packaging, and packing shall be as specified in MIL-E-17555 and MIL-STD-794, Level C. The method of preservation for Level C shall be determined in accordance with the selection chart in Appendix D specified in MIL-STD-794 and the method of packaging shall be as specified in MIL-P-116.

5.2 Marking.

5.2.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.2.2 Normal marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3 Spare parts. When specified (see 6.2.1), spare parts shall be preserved and packaged to the same specifications as a complete amplifier product except that weight of spare parts will determine selection of style of the box or container.

6. NOTES

6.1 Intended use. The AM-7157/DKT-59 amplifier-mixer is intended for use in the AN/DKT-59 Telemetry Data Transmitting Set.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title, and data of this specification.

b. Number of amplifiers to be procured.

c. First article inspection if different from 4.4.

d. Test plans and procedures if different from 4.1.2.

e. Facility assigned to perform inspections (see 4.1).

f. Test data and reports if different from 4.1.3.

g. Acceptance criteria if different from 4.4.1 and 4.5.1.

h. Inspection conditions if different from 4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

i. Test equipment accuracy if different from 4.3.2.

j. Level of packing and packaging if different from 5.1.

k. Special marking (see 5.2.1).

l. Spare parts preservation and packaging (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph number	Data requirements	Applicable DID number	Options
4.1.2	Procedures, Test	UDI-T-21347	----
4.1.3	Reports, Test	DI-T-2072	----
4.1.3.1	Electromagnetic Interference Test Report	DI-R-7062	----
4.1.4	Report, Failed Item Analysis	DI-R-7039	----
4.1.5	Reports, Reliability Test and Demonstration	DI-R-7034	----
4.3.1	Equipment Calibration Procedures	DI-R-7065	----
4.4.2	Failure Analysis and Corrective Action Report (First Article)	DI-R-5299	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the document's first article.

Preparing activity:
Navy - AS

(Project 5815-N117)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85652(AS)	Amplifier-Mixer AM-7157/DKT-59

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department
(ESSD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(ESSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.